

EXPERIMENT. 4. Kronig Penney Model

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ECA3102-NANO ELECTRONICS

Slot-c

Aim:

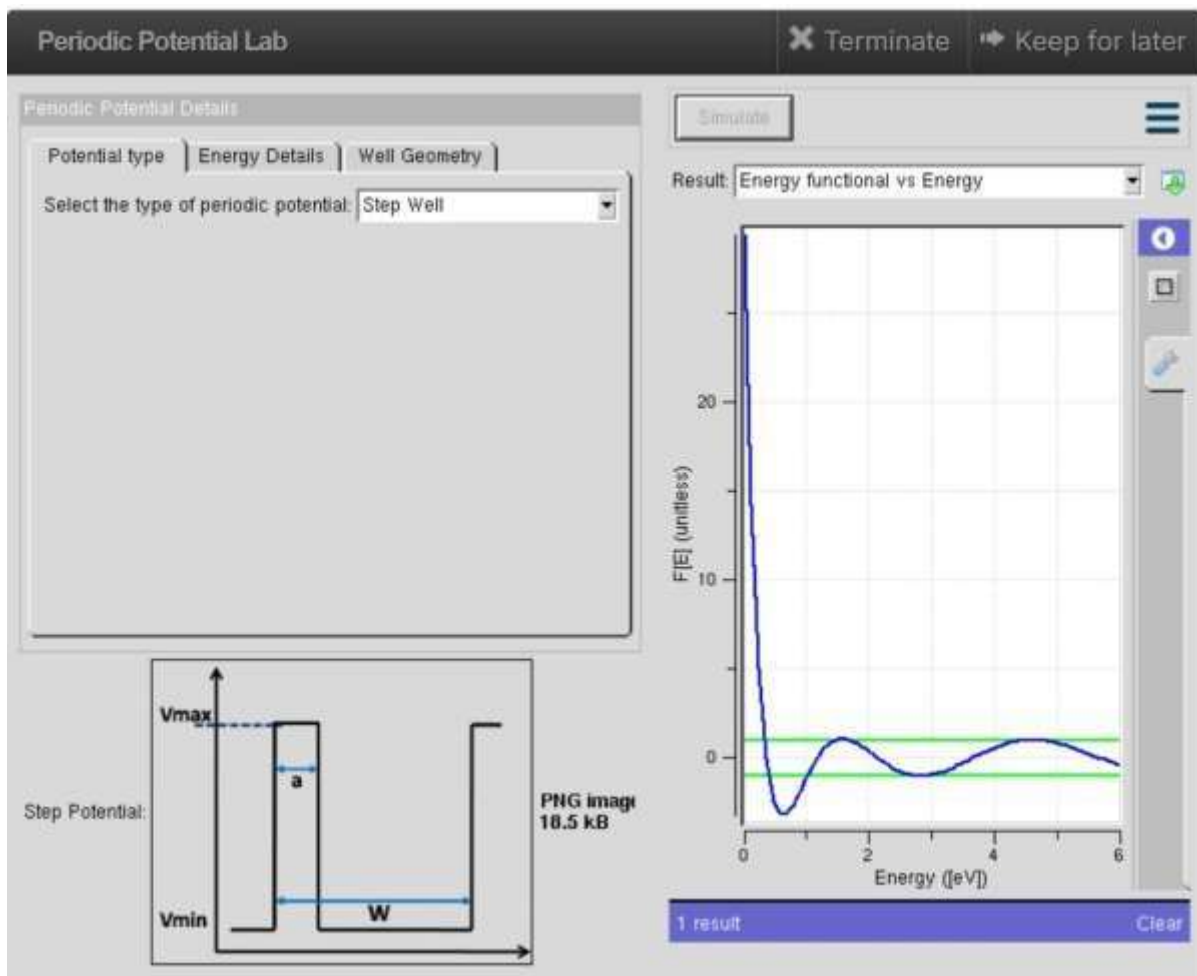
To simulate Energy functional vs Energy graph using Kronig Penney model

Software: NanoHub Online Tool

Procedure:

- Open nanoHUB. • Goto Resources -> Tools. • Select Kronig Penney Model in Tags. • Select Kronig Penney Model -> Periodic Potential Lab
- Press Launch tool in info section. • Select Potential Type-> Step well
- Select Energy details -> Maximum Barrier Height ->1eV
- Press simulate to view results. • Select Result - > Energy Functional Vs Energy

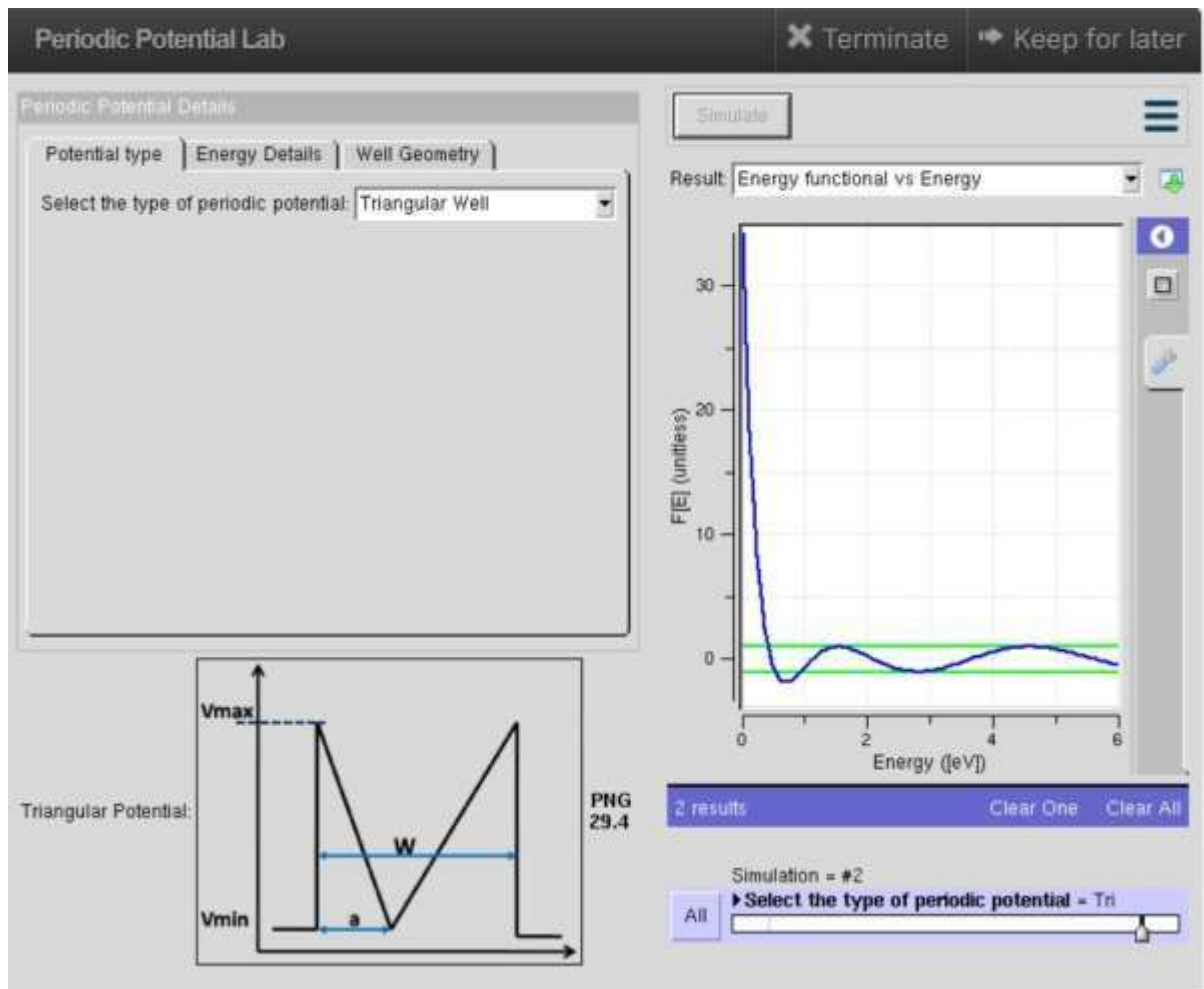
OUTPUT:



This figure shows the **Periodic Potential Lab** simulation using a **Step Well periodic potential**.

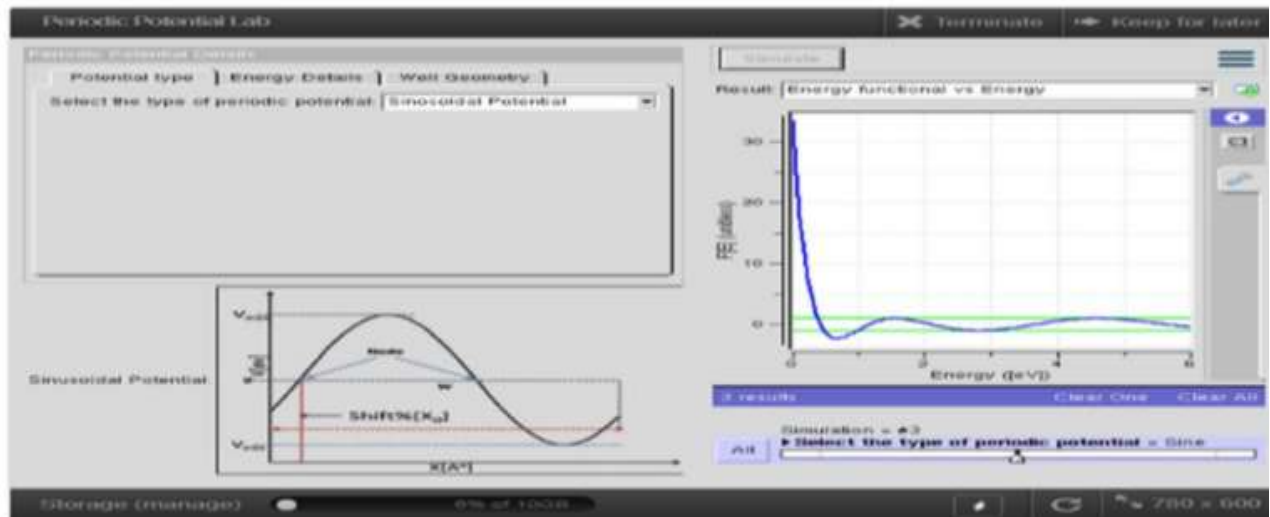
The graph illustrates the **Energy Functional vs. Energy**, showing how the electron energy varies within the periodic potential structure.

Case Study 1: Simulation of the output for Triangular well



This figure shows the **Periodic Potential Lab** simulation using a **Triangular Well periodic potential**. The graph shows the **Energy Functional vs. Energy**, illustrating how electron energy varies in the triangular potential well.

Case Study 2: Simulate the output for Sinusoidal Potential



"This simulation uses a sinusoidal periodic potential to study electron transport. The graph shows how the energy functional changes with electron energy."

Conclusion:

The simulations show that different periodic potential wells (step, triangular, and sinusoidal) affect the electron energy distribution and density of states. These results help analyze and optimize the electronic properties of nanostructured materials for nanoelectronic and thermoelectric applications.